EXPLAINABLE AI

UNIT 3- BANANA PROBLEM

TASK 3 AND 4 (EVEN SRN)

Name: NIHARIKA SAHA

SRN: PES1UG23AM190

Sec: AIML-D

Task 3: Feature Map Analysis

A pretrained ResNet18 model was used to analyze the banana image. The image was passed through the network, and feature maps were extracted from layer4. The channel with the highest average activation was identified and visualized.

The feature map highlights important regions of the image, showing that the model focuses on the curved structure and texture of the banana. This helps in understanding how the model internally represents visual features.

Feature Map Shape

[5]
✓ 0s

```
features = []

def hook_fn(module, input, output):
    features.append(output)

model.layer4.register_forward_hook(hook_fn)

output = model(input_tensor)
feature_map = features[0].detach()

print(feature_map.shape)
```

```
torch.Size([1, 512, 7, 7])
```

[6]
✓ 0s

```
avg_activation = feature_map.mean(dim=(2,3))
max_channel = avg_activation.argmax().item()
print("Max channel:", max_channel)
```

Most Activated Feature Map

[6]
✓ 0s

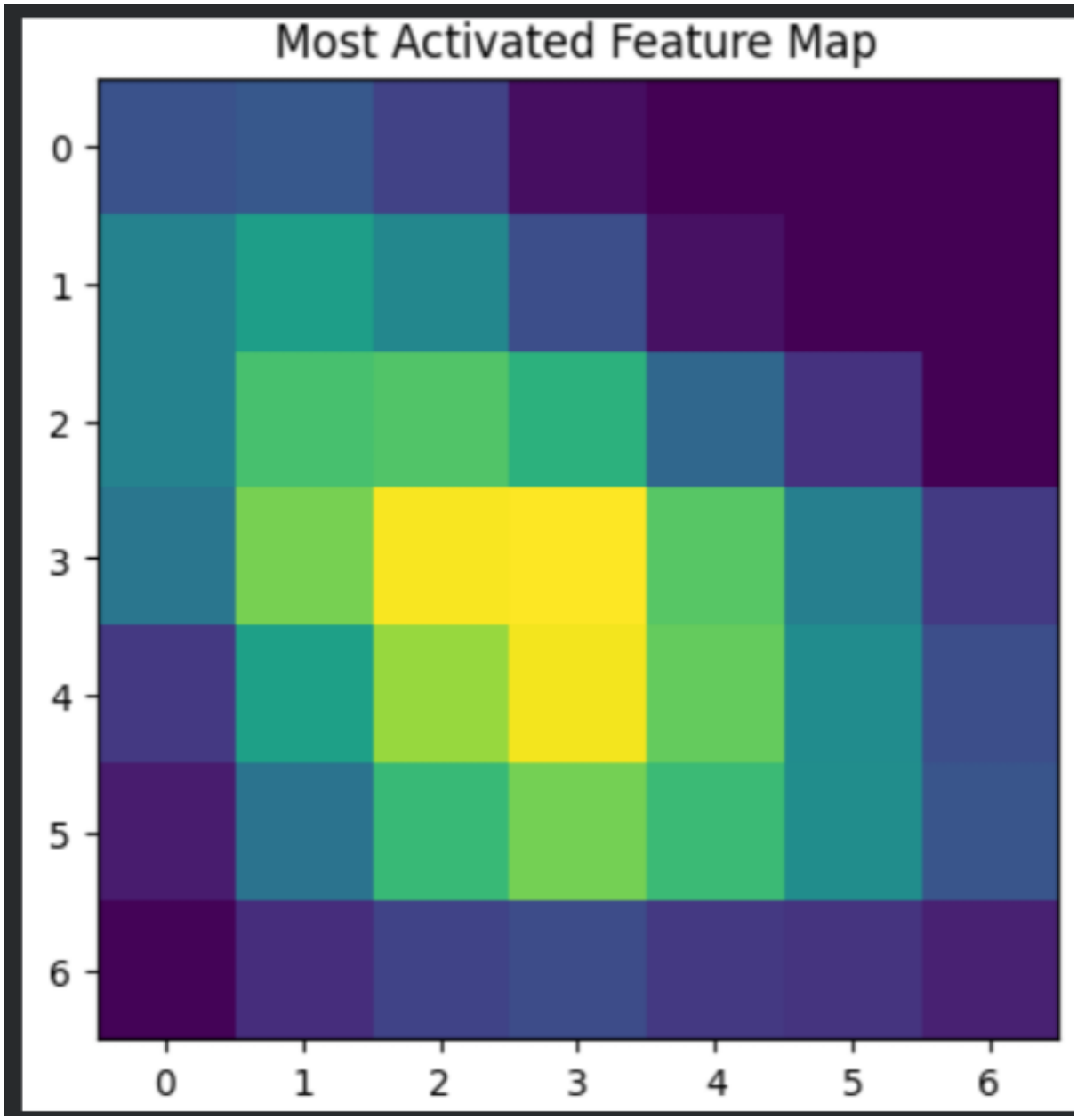
```
avg_activation = feature_map.mean(dim=(2,3))
max_channel = avg_activation.argmax().item()
print("Max channel:", max_channel)
```

```
Max channel: 499
```

[7]
✓ 0s

```
fm = feature_map[0, max_channel].cpu()

plt.imshow(fm, cmap='viridis')
plt.colorbar()
plt.title("Most Activated Feature Map")
plt.show()
```



Task 3 Observations

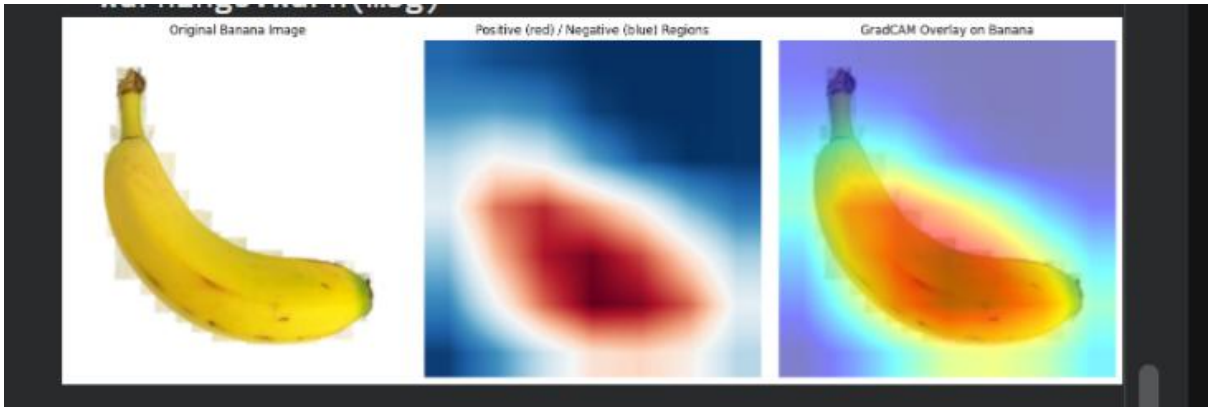
Some feature channels respond strongly to curved shapes, while others respond to color information such as the yellow surface of the banana. These feature maps help interpret the internal representations learned by the model.

Task 4: SHAP Explanation

SHAP (SHapley Additive exPlanations) was used to interpret the model's prediction for the banana image. The visualization shows the contribution of different regions of the image towards the prediction.

Red regions indicate positive contributions, while blue regions indicate negative contributions. The model primarily focuses on the yellow curved body of the banana, indicating that these features are most important for classification.

SHAP Visualization



Task 4 Observations

Red regions indicate positive contribution towards the prediction, while blue regions indicate negative contribution. SHAP provides more stable and interpretable explanations compared to basic gradient-based methods.